Load flows

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What is a load flow?

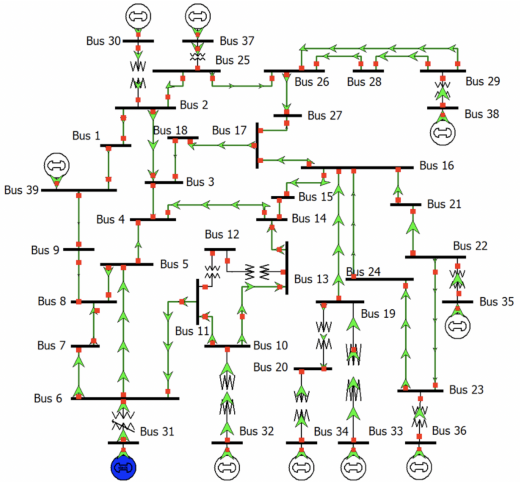
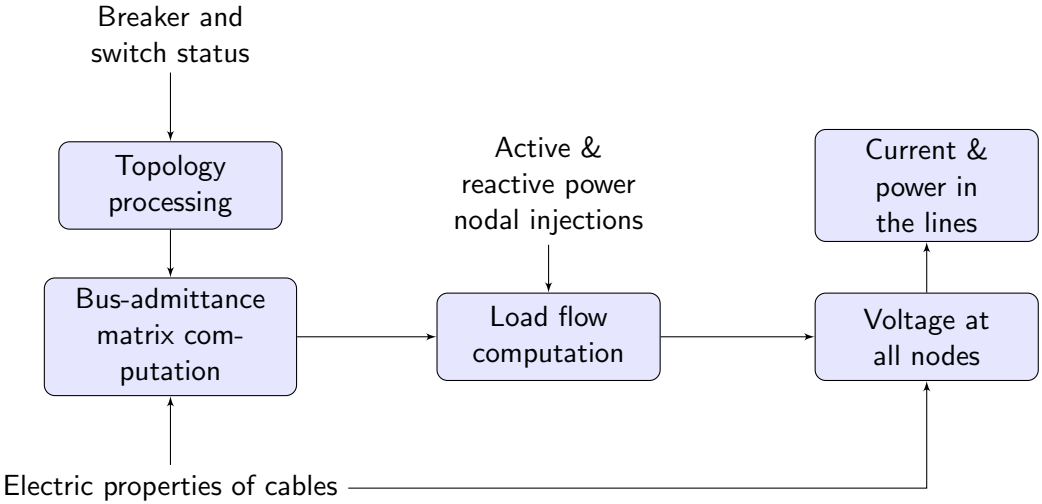


Diagram of a grid with 39 buses

The load flow problem is used to determine the voltages at all nodes of a network starting from the power injections at the nodes.

What is a load flow? Cont'd



What is a load flow? Cont'd

- The LF is a fundamental tool for grid operators, used for both operations (eg, contingency analyses) and planning
- Reference textbook for (most of) this part is *Power System Analysis* by Grainger and Stevenson

Outline of this module

- Per-unit representation
- Voltage-to-current relationship in a circuit
- Bus admittance matrix
- Formulation of the load flow problem
- Newton method to solve equations
- Solving the load flow problem

Per-unit representation

Per-unit representation

Per-unit representation facilitates the interpretation of network quantities (voltages, currents, impedances, and power). For example

For

- single-phase systems; or
- 3-phase systems where "current" refers to **line current**, "voltage" to **voltage-to-neutral**, and "kilovoltamperes" refers to **kilovoltamperes per phase**,

the following definitions hold:

$$\text{Base current [A]} = \frac{\text{base kVA}_{1p}}{\text{base voltage kV}_{LN}}$$

$$\text{Base impedance, } [\Omega] = \frac{\text{base voltage, } V_{LN}}{\text{base current, A}}$$

$$\text{Base power [kW}_{1p}] = \text{base kVA}_{1p}$$

Per-unit representation of three-phase circuits

Information for three-phase circuits are typically given in terms of:

- line voltage V_{LL} (as opposed to phase voltage V_{LN})
- total power over the three phases.

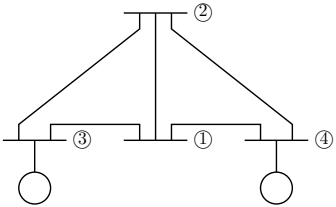
When computing per-unit bases in the former slide, V_{LL} should be divided by $\sqrt{3}$ and the total power by 3.

Power flows for 3-phase balanced circuits are investigated in terms of a single line-to-neutral circuit.

Voltage-to-current relationship in a circuit

Voltage-to-current model

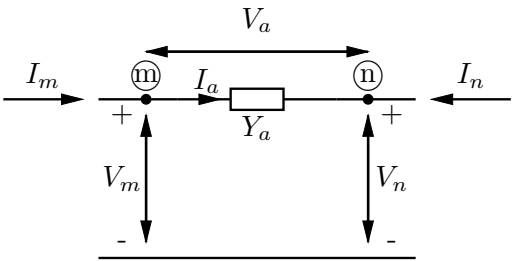
Consider a system with 2 AC generators, 4 lines, and 4 nodes (or buses):



The objective of the problem is to find voltage and current in the circuit as a function of the (known) injections of the generator and line parameters.

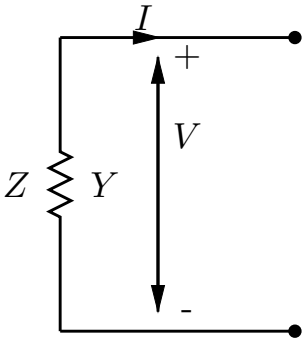
Simple line model and nomenclature

A line connecting two nodes can be modeled with the following branch model:



- Y_a is the branch admittance
- I_a is the branch current
- V_a is the voltage drop across the line
- m and n are the nodes
- V_m and V_n are the nodal voltages
- I_m and I_n are the current injected into the nodes

Admittance



Impedance

$$V = ZI$$

Impedance transforms from current to voltage.

Admittance

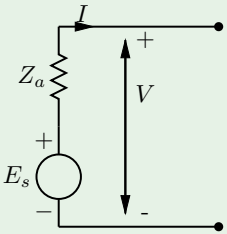
$$Y := \frac{1}{Z}$$

$$V = \frac{1}{Y} I \longrightarrow I = YV$$

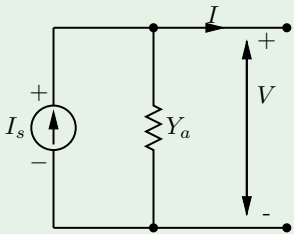
Admittance transforms from voltage to current.

Thevenin equivalent circuit

Generator connected to the network



Thevenin's equivalent

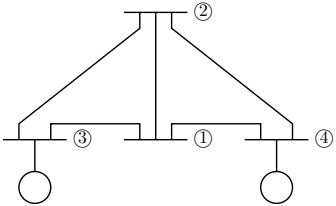


Kirchhoff voltage law	$E_s = IZ_a + V$
Dividing by the series impedance	$\frac{E_s}{Z_a} = I + V\frac{1}{Z_a}$
Generator with shunt impedance	$I_s = I + VY_a$

Voltage-to-current model: from single-line to admittance diagram

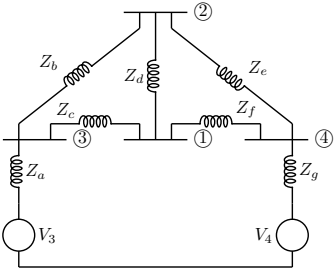
Step 0

Single line diagram



Step 1

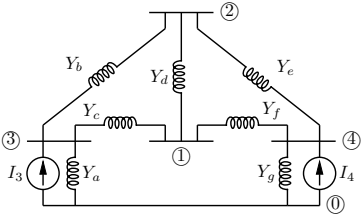
Impedance diagram



Lines modeled with reactances. Internal impendances of the generators combined with those of the lines

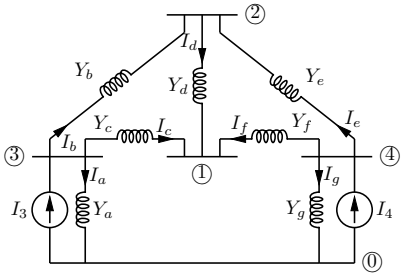
Step 2

Admittance diagram



Thevenin equivalents used to replace voltage sources with current sources

Voltage-to-current model: solving the circuit



Writing Kirchoff current law (*) at Node ① yields:

$$-I_c - I_d - I_f = 0$$
$$(V_1 - V_3)Y_c + (V_1 - V_2)Y_d + (V_1 - V_4)Y_f = 0$$

At Node ③:

$$I_a + I_b + I_c = I_3$$
$$V_3 Y_a + (V_3 - V_2)Y_b + (V_3 - V_1)Y_c = I_3$$

(*) Line currents away from the node equated to current into node.

Voltage-to-current model: solving the circuit (cont'd)

Re-arranging terms yields:

$$
\begin{aligned}
&① \quad V_1(Y_c + Y_d + Y_f) - V_2 Y_d - V_3 Y_c - V_4 Y_f = 0 \\
&③ \quad V_3(Y_a + Y_b + Y_c) - V_1 Y_c - V_2 Y_b = I_3
\end{aligned}
$$

For one node, the following rules can be deduced:

On the left-hand side of the equation:

- voltage at that node times the sum of the incident admittances **plus**
- the sum of the negative voltage of all nearby nodes times the impedance

For the right-hand side:

- current source entering the node;

Voltage-to-current model: solving the circuit (cont'd)

In matrix form, the circuit equations can be expressed in the following form:

$$
\begin{array}{c}
 \\ ① \\ ② \\ ③ \\ ④
\end{array}
\begin{array}{c}
\begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\
\begin{bmatrix}
Y_{11} & Y_{12} & Y_{13} & Y_{14} \\
Y_{21} & Y_{22} & Y_{23} & Y_{24} \\
Y_{31} & Y_{32} & Y_{33} & Y_{34} \\
Y_{41} & Y_{42} & Y_{43} & Y_{44}
\end{bmatrix}
\end{array}
\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}
=
\begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}
$$

where the row index i refers to the current leaving/entering node i, and the column index j refers to the contribution of voltage at node j to that current.

Voltage-to-current model: solving the circuit (cont'd)

$$Y_{\text{bus}} = \begin{bmatrix} Y_{11} & Y_{12} & Y_{13} & Y_{14} \\ Y_{21} & Y_{22} & Y_{23} & Y_{24} \\ Y_{31} & Y_{32} & Y_{33} & Y_{34} \\ Y_{41} & Y_{42} & Y_{43} & Y_{44} \end{bmatrix}, \quad Y_{\text{bus}} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

Y_{bus} is called the **bus admittance matrix** (or compound admittance matrix). Its terms are as follows:

- the diagonal element Y_{ii} equals the sum of the admittances directly connected to node i. These are called self-admittances at the nodes;
- off-diagonal element Y_{ij} equals the negative of the net admittance connected between nodes i and j; these are called mutual admittances of the nodes.

Bus admittance matrix

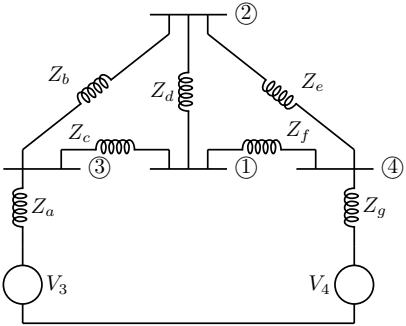
The bus admittance matrix Y_{bus} and relates the nodal current injections with the nodal voltages in a circuit.

Y_{bus} "condenses" all the topological information and electrical characteristics of a circuit and is an essential element to solve a load flow problem.

Self-Assessment of Understanding

1. Explain what the load flow problem does, including specifying inputs, outputs, and their nature (i.e., scalar values, complex values, etc).
2. State the (two: one for diagonal terms, one for off-diagonal terms) rules to compose the bus-admittance matrix Y.
3. Explain why the admittance matrix is symmetric.
4. What's the size of the bus admittance matrix?
5. In the relation $Y_{\text{bus}} V = I$, explain what are V and I, and explain the meaning of the equation.
6. Develop Exercise 1 of the next slide.

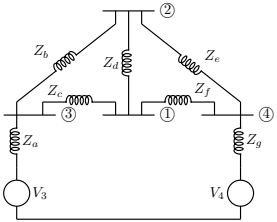
Exercise 1

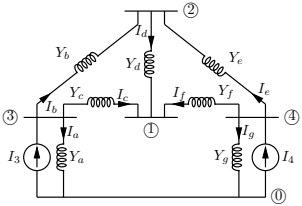


$$V_3 = 1.25\angle 0^\circ$$
$$V_4 = 0.85\angle -45^\circ$$
$$Z_a = j1.25$$
$$Z_b = j0.25$$
$$Z_c = j0.25$$
$$Z_d = j0.125$$
$$Z_e = j0.2$$
$$Z_f = j0.4$$
$$Z_g = j1.25$$

1. Compute the voltage magnitude at all nodes using the method seen above.
2. Implement the circuit in a circuit solver of your choice (e.g. Matlab Simulink) and compare the result.

Solution of Exercise 1



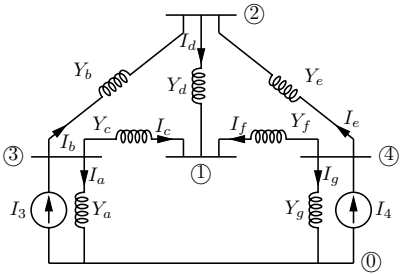


1. Chose a direction for the currents.
2. Compute Thevenin equivalent for voltage generators.
   - Generator 3: $Y_a = \dfrac{1}{Z_a}$, $I_3 = \dfrac{V_3}{Z_a}$
   - Generator 4: $Y_g = \dfrac{1}{Z_g}$, $I_4 = \dfrac{V_4}{Z_g}$
3. Compute branch admittances:

$$Y_x = \frac{1}{Z_x}, \quad \text{for } x = \{b, c, d, f, e\}$$

4. Compute the bus admittance matrix.
5. Solve $Y_{\text{bus}} V = I$ for V.

Solution of Exercise 1



$$I_3 = 1.00\angle{-90.0^\circ}$$
$$I_4 = 0.68\angle{-135.0^\circ}$$
$$Y_a = 0.00 - j0.80$$
$$Y_b = 0.00 - j4.00$$
$$Y_c = 0.00 - j4.00$$
$$Y_d = 0.00 - j8.00$$
$$Y_e = 0.00 - j5.00$$
$$Y_f = 0.00 - j2.50$$
$$Y_g = 0.00 - j0.80$$

Solution of Exercise 1

```
Y_bus =
  -14.5000    8.0000    8.0000    2.5000
    8.0000  -17.0000    4.0000    5.0000
    4.0000    4.0000   -8.8000         0
    2.5000    5.0000         0   -8.3000
```

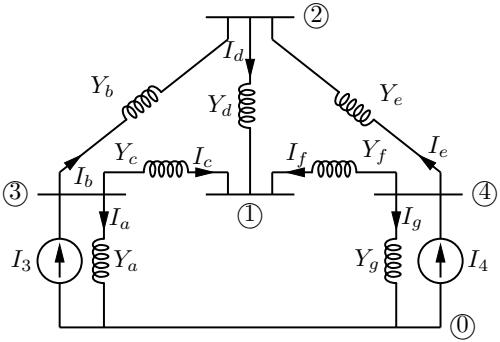
```
I =
   0.0000 + 0.0000i
   0.0000 + 0.0000i
   0.0000 + 1.0000i
  -0.4808 + 0.4808i
```

Voltage at the buses 1 – 4:

```
abs(inv(Y_bus) * I) =
    0.9030
    0.7752
    0.6529
    0.6706
```

A systematic way to produce Y_{bus}

Let us consider the former network:



By (arbitrarily) choosing a direction for the currents, we can create a direct graph connecting the nodes. The direction of the edges denotes the sense of the current.

The branch-to-node matrix

ψ
ϕ
θ

We create a matrix A, called **branch-to-node** matrix, with as many rows as number of branches (lines) and number of columns as number of nodes.

Each element a_{ij} of matrix A is defined as:

$$a_{ij} = \begin{cases} 1 & \text{if line } i \text{ departs from node } j \\ -1 & \text{if line } i \text{ arrives to node } i \\ 0 & \text{otherwise} \end{cases} .$$

(Node 0 is excluded)

$$A = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array}$$

The branch-to-node matrix (cont'd)

Multiplying A by the nodal voltages ...

$$
\begin{array}{c}
\\ a \\ b \\ c \\ d \\ e \\ f \\ g
\end{array}
\begin{array}{c}
\begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\
\begin{bmatrix}
0 & 0 & 1 & 0 \\
0 & -1 & 1 & 0 \\
-1 & 0 & 1 & 0 \\
-1 & 1 & 0 & 0 \\
0 & -1 & 0 & 1 \\
-1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
\underbrace{\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}}_{V}
=
\begin{bmatrix}
V_3 \\
-V_2 + V_3 \\
-V_1 + V3 \\
-V_1 + V2 \\
-V_2 + V4 \\
-V_1 + V4 \\
V4
\end{bmatrix}
=
\underbrace{\begin{bmatrix} V_a \\ V_b \\ V_c \\ V_d \\ V_e \\ V_f \\ V_f \end{bmatrix}}_{V_{\text{pr}}}
$$

... yields the voltage drops along the lines!

The branch-to-node matrix (cont'd)

We collect the **nodal voltages** in the vector V and the voltage drops along the lines in the vector V_{pr}.

$$
\begin{array}{c}
 \\ a \\ b \\ c \\ d \\ e \\ f \\ g
\end{array}
\begin{array}{c}
\begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\
\begin{bmatrix}
0 & 0 & 1 & 0 \\
0 & -1 & 1 & 0 \\
-1 & 0 & 1 & 0 \\
-1 & 1 & 0 & 0 \\
0 & -1 & 0 & 1 \\
-1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
\underbrace{\begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix}}_{V}
=
\underbrace{\begin{bmatrix} V_a \\ V_b \\ V_c \\ V_d \\ V_e \\ V_f \\ V_f \end{bmatrix}}_{V_{\text{pr}}}
$$

We say that:

$$V_{\text{pr}} = AV.$$

A closer look to the branch-to-node matrix

The columns of A tells the branches entering into the nodes, thus possibly providing information about currents entering and currents leaving...

$$
\begin{array}{c}
 \\ a \\ b \\ c \\ d \\ e \\ f \\ g
\end{array}
\begin{array}{c}
\begin{array}{cccc} ① & ② & ③ & ④ \end{array} \\
\begin{bmatrix}
0 & 0 & 1 & 0 \\
0 & -1 & 1 & 0 \\
-1 & 0 & 1 & 0 \\
-1 & 1 & 0 & 0 \\
0 & -1 & 0 & 1 \\
-1 & 0 & 0 & 1 \\
0 & 0 & 0 & 1
\end{bmatrix}
\end{array}
$$

A closer look to the branch-to-node matrix

If we group all branch currents into a column vector:

$$\begin{bmatrix} I_a & I_b & I_c & I_d & I_e & I_f & I_g \end{bmatrix}^T$$

and pre-multiply it by the transpose of A, then we get the current balance at the nodes:

$$\begin{array}{c} ① \\ ② \\ ③ \\ ④ \end{array} \begin{bmatrix} 0 & 0 & -1 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \\ I_d \\ I_e \\ I_f \\ I_g \end{bmatrix}$$

The branch-to-node matrix (cont'd)

The current balance at each should match the current nodal injections (zero if there are none - Kirchhoff current):

$$\begin{bmatrix} 0 & 0 & -1 & -1 & 0 & -1 & 0 \\ 0 & -1 & 0 & 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} I_a \\ I_b \\ I_c \\ I_d \\ I_e \\ I_f \\ I_g \end{bmatrix}}_{I_{pr}} = \underbrace{\begin{bmatrix} 0 \\ 0 \\ I_3 \\ I_4 \end{bmatrix}}_{I} \rightarrow A^\top I_{\text{pr}} = I$$

With the definition of I_{pr} and I in place, then:

$$A^T I_{pr} = I.$$

Primitive admittance matrix

We now introduce the **primitive admittance matrix** that simply relates branch voltages and branch currents:

$$\begin{bmatrix} Y_a & & \\ & \ddots & \\ & & Y_f \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \\ V_d \\ V_e \\ V_f \\ V_f \end{bmatrix} = \begin{bmatrix} I_a \\ I_b \\ I_c \\ I_d \\ I_e \\ I_f \\ I_g \end{bmatrix}.$$

It is a block diagonal matrix composed of all primitive impedances of the individual lines.

A systematic way to produce the Y-bus matrix (cont'd)

Parts we have:

$$V_{\text{pr}} = AV \qquad (1)$$

$$A^\top I_{pr} = I \qquad (2)$$

$$Y_{\text{pr}} V_{\text{pr}} = I_{\text{pr}} \qquad (3)$$

Pre-multiplying both sides of (3) by $A^\top$ yields

$$A^\top Y_{\text{pr}} V_{\text{pr}} = A_{\text{pr}}^\top I_{pr} \;\rightarrow\; A^\top Y_{\text{pr}} V_{\text{pr}} = I.$$

Replacing V_{pr} with (1) above gives:

$$A^\top Y_{\text{pr}} AV = I. \qquad (4)$$

Define $Y_{\text{bus}} = A^\top Y_{\text{pr}} A$, then (4) becomes:

$$Y_{\text{bus}} V = I.$$

Bus admittance matrix: Summary

- We have seen how to build the bus admittance matrix for a circuit, first reasoning on the circuit, then using an automated procedure more suitable for computer calculations.
- The bus admittance matrix transforms the voltage of a circuit into the current injections in the nodes:

$$Y_{\text{bus}} V = I$$

- Inverting the linear equation above would allow us to answer the question: “”What are the nodal voltages as a function of the nodal current injections?””
- Because we rather reason in terms of power injections than in current injections, it is more convenient to formulate the question in terms of: “”What are the nodal voltages as a function of the nodal power injections?””

Formulation of the load flow problem

Complex power

Given with the following voltage and current phasor:

$$V = v\underline{/\phi}$$
$$I = i\underline{/\psi},$$

the associated complex power is given by

$$\begin{aligned} VI^* &= vi\underline{/\phi - \psi} = \\ &= vi\left(\cos(\phi - \psi) + j\sin(\phi - \psi)\right) = \\ &= P + jQ := S \end{aligned}$$

Load flow equations

The power injection at a node (j) can be expressed as:

$$P_j - jQ_j = V_j^* I_j.$$

Given with the bus admittance matrix Y_{bus} (Y for brevity), the nodal injected currents are:

$$I = YV$$

The current at node j is the j-th element of the product $I = YV$, so:

$$P_j - jQ_j = V_j^* \sum_{n=1}^{N} Y_{jn} V_n.$$

Load flow equations: nomenclature

- The complex voltage (or voltage phasor) at the node is

$$V_i = v_i \underline{/\phi_i}$$

where v_i is the voltage magnitude, and ϕ_i is the voltage phase angle.

- The complex impedance is

$$Y_{jn} = y_{jn} \underline{/\theta_i} = G_{jn} + jB_{jn}$$

where G is the conductance and B the susceptance.

Load flow equations (cont'd)

$$P_j - jQ_j = V_j^* \sum_{n=1}^{N} Y_{jn} V_n = \qquad \text{(See note)}$$

$$= v_j \sum_{n=1} y_{jn} v_n \cos(\theta_{jn} + \phi_n - \phi_j) + j v_j \sum_{n=1} y_{jn} v_n \sin(\theta_{jn} + \phi_n - \phi_j).$$

Splitting into active and reactive power yields:

$$P_j = +v_j \sum_{n=1} y_{jn} v_n \cos(\theta_{jn} + \phi_n - \phi_j)$$

$$Q_j = -v_j \sum_{n=1} y_{jn} v_n \sin(\theta_{jn} + \phi_n - \phi_j).$$

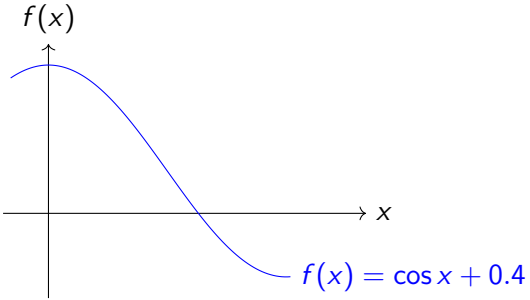
Note: product of complex numbers in polar form

$$A := a\underline{/\alpha}, B := b\underline{/\beta}, C := c\underline{/\gamma}, \quad \text{then:} \quad A^* \cdot B \cdot C = a \cdot b \cdot c\underline{/-\alpha + \beta + \gamma}$$

Newton method for solving functions

Solving an equation

Let us consider the following function $f(x) = \cos x + 0.4$ with domain $0 \leq x \leq 3.2$:

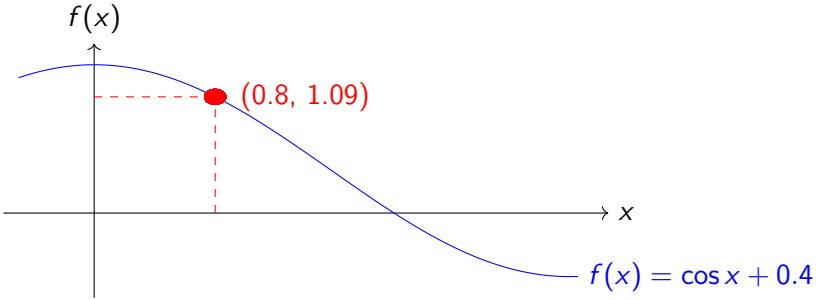


We want to find a solution of the equation, namely finding a value of the function argument $x^\star$ such that the function evaluated at that point is zero, $f(x^\star) = 0$.

Solving an equation: a rough initial solution

Assume someone provides us with an initial guess, $x_x^{(0)}$, for the solution::

$$x_0^\star = 0.8; \;\; f\left(x_0^\star\right) = 1.09 \qquad\qquad \text{(Is this a good guess?)}$$



Can we make a better guess?

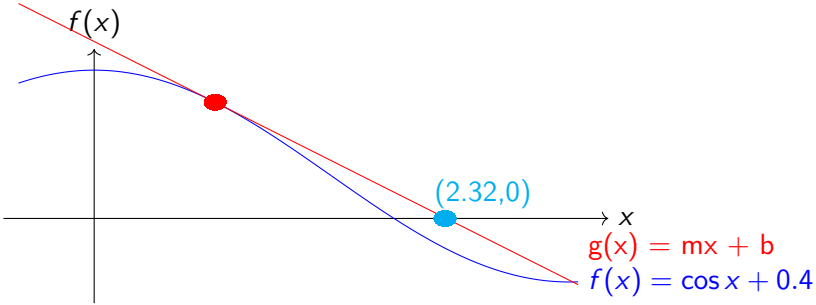
Solving an equation: a better guess?

We can approximate the function $f(x)$ at the point $x_0^\star$ with a linear function

$$g(x) = mx + b.$$

Because $g(x)$ is linear, we can solve $g(x) = 0$ easily. Its solution is given by:

$$g(x) = 0 \longrightarrow mx + b = 0 \longrightarrow x = \frac{-b}{m}$$



Solving an equation: a better guess (cont'd)?

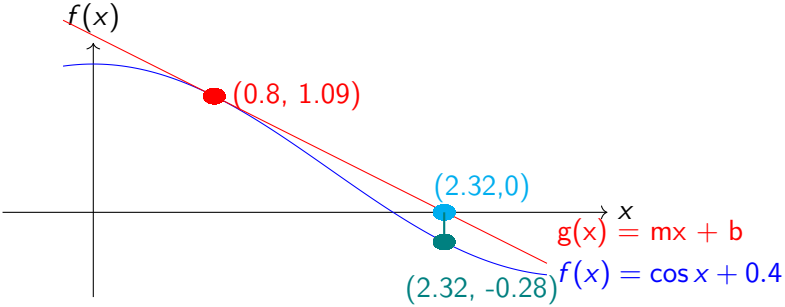
Is the guessed solution found with the linear approximation:

$$x_1^\star = -b/m = 2.32$$

better than the former guess $x_0^\star$? The value of $f(x)$ at the new guess is:

$$f(x_1^\star) = -0.28.$$

Because -0.28 is closer to 0 than 1.09, $x_1^\star = 2.32$ is a better guess than $x_0^\star = 0.8$!



Solving an equation: Iterations

We can linearize again around the new point (2.32,-0.28) and repeat the procedure.

If we do so, we obtain:

Iteration i	$x_i^\star$	$f(x_i^\star)$
0	0.8	1.0967
1	2.3288	-0.2875
2	1.9329	0.0457
3	1.9818	0.0004
4	1.9823	0
5	1.9823	0

We can stop at a fixed number of iterations or when the $f(x_i^\star)$ is reasonably close to 0.

Solving an equation: summary of the Newton method

1. Store a guessed solution in $x_0^\star$, initialize iteration index $i = 0$, store a (small) tolerance value for ending the iterations in ϵ.
2. Compute linear approximation at $x_i^\star$ (1st order Taylor expansion):

$$\tilde{f}(x) = f(x_i^\star) + \underbrace{\dot{f}(x_i^\star)}_{\text{We call this } m} (x - x_i^\star)$$

   where $\dot{f}(x_i^\star) = \left.\frac{df}{dx}\right|_{x=x_i^\star}$ is the first derivative of $f(x)$ calculated at $x_i^\star$ (slope).
3. Find where the linear approximation crosses zero (set it to zero, and solve for x); this is our new estimate of the solution:

$$0 = f(x_i^\star) + mx - mx_i^\star \quad \longrightarrow \quad x = x_i^\star - \frac{f(x_i^\star)}{m} \quad \longrightarrow \quad x_{i+1}^\star = x_i^\star - \frac{f(x_i^\star)}{m}.$$

4. Is $f(x_{i+1}^\star) \leq \epsilon$? No: $i = i + 1$ and go to Step 2. Yes: exit.

Exercise 3

- Write your own Newton solver for equations in your favourite programming language. A code template is available.
- Reproduce and verify the result of the former table. Function: $f(x) = \cos x + 0.4$; domain: $0 \leq x \leq 3.2$. (why is it necessary to specify a domain?).
- Find the point (x, y) where the function

$$f(x) = e^{-0.1x}$$

crosses the y axis at value 0.4.

Newton method: extension to multiple variables

We consider two functions

$$g_1(x, y) = 0$$
$$g_2(x, y) = 0$$

where x and y are independent variables and u is a parameter (e.g., some constant input value).

Extension to multiple variables (cont'd)

The linear approximations (1st order Taylor expansion) of f and g at $x_0^\star, y_0^\star$ is:

$$\tilde{g}_1(x,y) = g_1(x_0^\star, y_0^\star) + \left.\frac{\delta g_1}{\delta x}\right|_{x_0^\star, y_0^\star}(x - x_0^\star) + \left.\frac{\delta g_1}{\delta y}\right|_{x_0^\star, y_0^\star}(y - y_0^\star)$$

$$\tilde{g}_2(x,y) = g_2(x_0^\star, y_0^\star) + \left.\frac{\delta g_2}{\delta x}\right|_{x_0^\star, y_0^\star}(x - x_0^\star) + \left.\frac{\delta g_2}{\delta y}\right|_{x_0^\star, y_0^\star}(y - y_0^\star)$$

For compactness, we define:

$$\Delta y := y - y_0^\star, \quad \Delta x := x - x_0^\star.$$

The former expressions become:

$$\tilde{g}_1(x,y) = g_1(x_0^\star, y_0^\star) + \left.\frac{\delta g_1}{\delta x}\right|_{x_0^\star, y_0^\star}\Delta x + \left.\frac{\delta g_1}{\delta y}\right|_{x_0^\star, y_0^\star}\Delta y$$

$$\tilde{g}_2(x,y) = g_2(x_0^\star, y_0^\star) + \left.\frac{\delta g_2}{\delta x}\right|_{x_0^\star, y_0^\star}\Delta x + \left.\frac{\delta g_2}{\delta y}\right|_{x_0^\star, y_0^\star}\Delta y.$$

Extension to multiple variables (cont'd)

In the spirit of Newton's method, we equate the linear approximations to zero so that we can solve the system of equations for $\Delta x, \Delta y$:

$$\left.\frac{\delta g_1}{\delta x}\right|_{x_0^\star, y_0^\star} \Delta x + \left.\frac{\delta g_1}{\delta y}\right|_{x_0^\star, y_0^\star} \Delta y = -g_1(x_0^\star, y_0^\star)$$
$$\frac{\delta g_2}{\delta x}\Big|_{x_0^\star, y_0^\star} \Delta x + \frac{\delta g_2}{\delta y}\Big|_{x_0^\star, y_0^\star} \Delta y = -g_2(x_0^\star, y_0^\star).$$

In (a more convenient) matrix notation:

$$\begin{bmatrix} \frac{\delta g_1}{\delta x} & \frac{\delta g_1}{\delta y} \\ \frac{\delta g_2}{\delta x} & \frac{\delta g_2}{\delta y} \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} -g_1(x_0^\star, y_0^\star) \\ -g_2(x_0^\star, y_0^\star) \end{bmatrix}$$

Extension to multiple variables (cont'd)

By introducing:

$$J := \underbrace{\begin{bmatrix} \frac{\delta g_1}{\delta x} & \frac{\delta g_1}{\delta y} \\ \frac{\delta g_2}{\delta x} & \frac{\delta g_2}{\delta y} \end{bmatrix}}_{\text{Jacobian matrix}}, \qquad \mathcal{X} := \underbrace{\begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}}_{\substack{\text{Problem's} \\ \text{unknowns}}}, \qquad \mathcal{G} := \underbrace{\begin{bmatrix} -g_1(x_0^\star, y_0^\star) \\ -g_2(x_0^\star, y_0^\star) \end{bmatrix}}_{\substack{\text{Function to solve} \\ \text{calculated at } (x_0^\star, y_0^\star)}},$$

the linear transformation seen before becomes:

$$J\mathcal{X} = \mathcal{G}.$$

Solving for the unknown X yields:

$$\mathcal{X} = J^{-1}\mathcal{G}.$$

Extension to multiple variables (cont'd)

Once

$$\mathcal{X} := \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

is computed, then the new estimated solution is found by carrying out the following update:

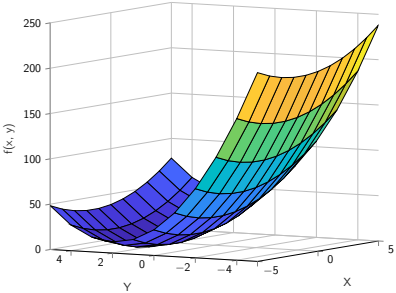
$$\begin{bmatrix} x_1^\star \\ y_1^\star \end{bmatrix} = \begin{bmatrix} x_0^\star \\ y_0^\star \end{bmatrix} + \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}.$$

This update is iterated as in the case of the single-variable function until an end criterion is met.

Exercise 4

Write a solver to solve the following equation:

$$(x + 1.5)^2 + (2y - 4)^2 = 0$$



Note that:

- the solution is trivial (you can compute it by eye), nevertheless the exercise is to write the code and validate it; and
- you have two variables, but one equation only; you need to slightly adapt the formulation seen before.

Solving the load flow equations

Load flow problem: definition of nodes

- Slack bus: the voltage of the slack bus is fixed by definition (e.g., 1 PU), and the power is free (not calculated). To solve a load flow, one has always to select one slack bus and assign a voltage to it.
- PQ nodes: these are nodes where there are nodal injections. The voltage of these nodes is the unknown of the problem.
- Voltage-controlled nodes: nodes where the voltage can be assumed fixed (simplifying the problem).

Recall of the load flow equations

Load flow equations (from former slides):

$$P_j = +v_j \sum_{n=1} y_{jn} v_n \cos(\theta_{jn} + \phi_n - \phi_j)$$
$$Q_j = -v_j \sum_{n=1} y_{jn} v_n \sin(\theta_{jn} + \phi_n - \phi_j).$$

For each node j, we have:

- two equations with known terms P_j and Q_j (active and reactive power injections at node j), and θ_{jn} (angle of the impedance); and
- two unknowns: v_j and ϕ_j (voltage magnitude and voltage angle at node j).

The unknown can be solved by applying Netwon's method for multiple variables.

Recall of the load flow equations (cont'd)

Load flow equations:

$$P_j = +v_j \sum_{n=1} y_{jn} v_n \cos(\theta_{jn} + \phi_n - \phi_j)$$
$$Q_j = -v_j \sum_{n=1} y_{jn} v_n \sin(\theta_{jn} + \phi_n - \phi_j).$$

for all nodes $j = 1, \ldots, J$. Re-arranging terms yields:

$$+v_j \sum_{n=1} y_{jn} v_n \cos(\theta_{jn} + \phi_n - \phi_j) - P_j = 0$$
$$-v_j \sum_{n=1} y_{jn} v_n \sin(\theta_{jn} + \phi_n - \phi_j) - Q_j = 0.$$

A

We introduce the function $h(\cdot)$ to write the load flow equations in a more compact way:

$$P_j = h_j^P(v_1, v_2, \dots, \phi_1, \phi_2, \dots) \tag{5}$$

$$Q_j = h_j^Q(v_1, v_2, \dots, \phi_1, \phi_2, \dots). \tag{6}$$

Solving the equations entails finding the zero of the following expression:

$$0 = \underbrace{h_j^P(v_1, v_2, \dots, \phi_1, \phi_2, \dots) - P_j}_{g_j^P(\cdot)}$$

$$0 = \underbrace{h_j^Q(v_1, v_2, \dots, \phi_1, \phi_2, \dots) - Q_j}_{g_j^Q(\cdot)}.$$

where the $\cdot$ notation refers to all the argument list. In short, for each node j we have:

$$g_j^P(\cdot) = 0$$

$$g_j^Q(\cdot) = 0.$$

Solving the load flow

We want to solve the load flows equations:

$$g_j^P(v_1, v_2, \ldots, v_N, \phi_1, \phi_2, \ldots, \phi_N) = 0$$
$$g_j^Q(v_1, v_2, \ldots, v_N, \phi_1, \phi_2, \ldots, \phi_N) = 0.$$

for all j.

This is a system of equations with $2 \times N$ equations and $2 \times N$ unknowns (voltage magnitude v_j and phase angle ϕ_j at each node j), where N is the number of nodes.

The input of this problem are the active and reactive power injection at each node of the grid, P_j and Q_j.

The load flow equations can be solved by applying the Newton method for multiple variables (recapped in the next slide).

Recap of Newton methods for multiple variables

We want to find the solution of the equations:

$$g_1(x, y) = 0$$
$$g_2(x, y) = 0$$

The solution is found by iterating the following update ("old" is replaced by "new" at each iteration):

$$\begin{bmatrix} x^\star_{\text{new}} \\ y^\star_{\text{new}} \end{bmatrix} = \begin{bmatrix} x^\star_{\text{old}} \\ y^\star_{\text{old}} \end{bmatrix} + \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}, \tag{7a}$$

where:

$$\begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = J^{-1} b \tag{7b}$$

$$J = \begin{bmatrix} \frac{\partial g_1}{\delta x} & \frac{\partial g_1}{\delta y} \\ \frac{\partial g_2}{\delta x} & \frac{\partial g_2}{\delta y} \end{bmatrix} \tag{7c}$$

$$b = \begin{bmatrix} -g_1(x^{\text{old}}, y^{\text{old}}) \\ -g_2(x^{\text{old}}, y^{\text{old}}) \end{bmatrix} \tag{7d}$$

Solving the load flow: Procedure

Using the Newton method to solve the load flow equations

$$g_j^P(\cdot) = 0$$
$$g_j^Q(\cdot) = 0$$

for all j, requires the following steps:

1. Store the initial guess of the solutions in the variables $(v_1^{\text{old}}, \dots, v_N^{\text{old}}, \phi_1^{\text{old}}, \dots, \phi_N^{\text{old}})$.
2. Calculate Jacobian matrix J with (7c) – see next slides for more.
3. Calculate the equations $g(\cdot)$ in (7d) at the guessed solutions – see next slides for more.
4. Compute correction Δ of the guessed solutions with (7b)
5. Compute new solutions $(v_1^{\text{new}}, \dots, v_N^{\text{new}}, \phi_1^{\text{new}}, \dots, \phi_N^{\text{new}})$ with (7a).
6. Replace old guess with the new $(v_1^{\text{old}} = v_1^{\text{new}}, \dots)$ and repeat from Step 2 until convergence.

Solving the load flow: Step 2 – Jacobian computation

The Jacobian matrix contains the derivative of all the functions $g_j^X(\cdot)$ (along the rows) with respect to all the unknowns (columns). Developing the derivative of $g_j^P(\cdot)$, we obtain:

$$\frac{\partial g_j^P(\cdot)}{\partial v_1} = \frac{\partial \left(h_j^P(\cdot) - P_j\right)}{\partial v_1}.$$

Because P_j is a constant (and assumed voltage independent), its derivative with respect to v_j is zero, thus the expression above becomes:

$$\frac{\partial \left(h_j^P(\cdot) - P_j\right)}{\partial v_1} = \frac{\partial h_j^P(\cdot)}{\partial v_1}.$$

The derivatives of functions $g(\cdot)$'s are ultimately the derivative of functions $h(\cdot)$'s with respect to all variables. By definition of $h_j^P(\cdot)$ in (5), $h_j^P(v_1^{\text{old}}, \ldots, v_N^{\text{old}}, \phi_1^{\text{old}}, \ldots, \phi_N^{\text{old}})$ computes the active power at node j at the guessed solution. We can thus interpret the derivative of $h_j^P(\cdot)$ and $h_j^Q(\cdot)$ as the derivatives of active and reactive power injections.

Solving the load flow: Step 2 – Jacobian computation (cont'd)

The Jacobian matrix for the load flow problem is as follows:

$$J = \begin{bmatrix} \begin{bmatrix} \frac{\partial P_1}{\partial v_1} & \cdots & \frac{\partial P_1}{\partial v_N} \\ \vdots & & \vdots \\ \frac{\partial P_N}{\partial v_1} & \cdots & \frac{\partial P_N}{\partial v_N} \end{bmatrix} & \begin{bmatrix} \frac{\partial P_1}{\partial \phi_1} & \cdots & \frac{\partial P_1}{\partial \phi_N} \\ \vdots & & \vdots \\ \frac{\partial P_N}{\partial \phi_1} & \cdots & \frac{\partial P_N}{\partial \phi_N} \end{bmatrix} \\ \begin{bmatrix} \frac{\partial Q_1}{\partial v_1} & \cdots & \frac{\partial Q_1}{\partial v_N} \\ \vdots & & \vdots \\ \frac{\partial Q_N}{\partial v_1} & \cdots & \frac{\partial Q_N}{\partial v_N} \end{bmatrix} & \begin{bmatrix} \frac{\partial Q_1}{\partial \phi_1} & \cdots & \frac{\partial Q_1}{\partial \phi_N} \\ \vdots & & \vdots \\ \frac{\partial Q_N}{\partial \phi_1} & \cdots & \frac{\partial Q_N}{\partial \phi_N} \end{bmatrix} \end{bmatrix}$$

The inverse of the Jacobian matrix is said to express the sensitivity of the voltage phasors in the network to variations of the power injections.

Solving the load flow: Step 3 – $g(\cdot)$ at the guessed solutions

The vector b in (7d) is composed by the negative values of the functions g calculated at the guessed solutions.

Developing, as an example, for the case of $-g_j^P(\cdot)$, we get:

$$-g_j^P(v_1^{\text{old}}, \ldots, v_N^{\text{old}}, \phi_1^{\text{old}}, \ldots, \phi_N^{\text{old}}) = -h_j^P(v_1^{\text{old}}, \ldots, v_N^{\text{old}}, \phi_1^{\text{old}}, \ldots, \phi_N^{\text{old}}) + P_j =$$
$$= P_j - h_j^P(\cdot)$$
$$= \text{“Observed - computed injections”}$$

By definition of $h_j^P(\cdot)$ in (5), $h_j^P(v_1^{\text{old}}, \ldots, v_N^{\text{old}}, \phi_1^{\text{old}}, \ldots, \phi_N^{\text{old}})$ computes the active power at node j at the guessed solution. We can thus interpret the expression $P_j - h_j^P(\cdot)$ as the “observed” nodal injection minus the “computed” nodal injections at the guessed solutions. At the solution, this difference should be near 0.

Solving the load flow: Step 3 – $g(\cdot)$ at the guessed solutions (cont'd)

The vector b has thus the following composition:

$$b = \begin{bmatrix} P_1 - h_1^P(\cdot) \\ \vdots \\ P_N - h_N^P(\cdot) \\ Q_1 - h_1^Q(\cdot) \\ \vdots \\ Q_N - h_N^Q(\cdot) \end{bmatrix} = \begin{bmatrix} \Delta P_1 \\ \vdots \\ \Delta P_N \\ \Delta Q_1 \\ \vdots \\ \Delta Q_N \end{bmatrix}$$

where the notation $(\cdot)$ refers to the list of arguments: $v_1^{\text{old}}, \ldots, v_N^{\text{old}}, \phi_1^{\text{old}}, \ldots, \phi_N^{\text{old}}$.

Slack bus

("Bus" is another word for "node")

- The load flow problem computes all voltage phasors in a network as a function of the nodal injections.
- However, it is practical (and necessary) to have one node to use as a voltage reference for all the other nodes of the network.
- This node is called "slack bus". By definition, the complex voltage of the slack bus is constant. The slack bus also provides any remaining power to the grid.
- The choice of the slack bus is arbitrary; however, it is convenient to choose it in a convenient way so that we can leverage the property of the slack bus (i.e., constant voltage): for example, in a radial grid the slack bus would be the node at the connection point with the upper-grid level where the voltage can be considered constant.
- Because the slack bus has constant voltage (magnitude and phase), the update of its value is not carried out in the Newton method. This is done by removing from the Jacobian matrix the row and column referring to the slack bus, as shown next.

Slack bus and modification to the Jacobian: Working Example

Example for a 3-bus system where node 1 is the slack bus.

$$
\begin{bmatrix} \cancel{v_1^{\text{new}}} \\ \cancel{\phi_1^{\text{new}}} \\ v_2^{\text{new}} \\ \phi_2^{\text{new}} \\ v_3^{\text{new}} \\ \phi_3^{\text{new}} \end{bmatrix}
=
\begin{bmatrix} \cancel{v_1^{\text{old}}} \\ \cancel{\phi_1^{\text{old}}} \\ v_2^{\text{old}} \\ \phi_2^{\text{old}} \\ v_3^{\text{old}} \\ \phi_3^{\text{old}} \end{bmatrix}
+
\begin{bmatrix}
\cancel{\frac{\partial P_1}{\partial v_1}} & \cancel{\frac{\partial P_1}{\partial \phi_1}} & \cancel{\frac{\partial P_1}{\partial v_2}} & \cancel{\frac{\partial P_1}{\partial \phi_2}} & \cancel{\frac{\partial P_1}{\partial v_3}} & \cancel{\frac{\partial P_1}{\partial \phi_3}} \\
\cancel{\frac{\partial P_2}{\partial v_1}} & \cancel{\frac{\partial P_2}{\partial \phi_1}} & \frac{\partial P_2}{\partial v_2} & \frac{\partial P_2}{\partial \phi_2} & \frac{\partial P_2}{\partial v_3} & \frac{\partial P_2}{\partial \phi_3} \\
\cancel{\frac{\partial P_3}{\partial v_1}} & \cancel{\frac{\partial P_3}{\partial \phi_1}} & \frac{\partial P_3}{\partial v_2} & \frac{\partial P_3}{\partial \phi_2} & \frac{\partial P_3}{\partial v_3} & \frac{\partial P_3}{\partial \phi_3} \\
\cancel{\frac{\partial Q_1}{\partial v_1}} & \cancel{\frac{\partial Q_1}{\partial \phi_1}} & \cancel{\frac{\partial Q_1}{\partial v_2}} & \cancel{\frac{\partial Q_1}{\partial \phi_2}} & \cancel{\frac{\partial Q_1}{\partial v_3}} & \cancel{\frac{\partial Q_1}{\partial \phi_3}} \\
\cancel{\frac{\partial Q_2}{\partial v_1}} & \cancel{\frac{\partial Q_2}{\partial \phi_1}} & \frac{\partial Q_2}{\partial v_2} & \frac{\partial Q_2}{\partial \phi_2} & \frac{\partial Q_2}{\partial v_3} & \frac{\partial Q_2}{\partial \phi_3} \\
\cancel{\frac{\partial Q_3}{\partial v_1}} & \cancel{\frac{\partial Q_3}{\partial \phi_1}} & \frac{\partial Q_3}{\partial v_2} & \frac{\partial Q_3}{\partial \phi_2} & \frac{\partial Q_3}{\partial v_3} & \frac{\partial Q_3}{\partial \phi_3}
\end{bmatrix}^{-1}
\begin{bmatrix} \cancel{\Delta P_1} \\ \Delta P_2 \\ \Delta P_3 \\ \cancel{\Delta Q_1} \\ \Delta Q_2 \\ \Delta Q_3 \end{bmatrix}
$$

Cancelled terms, $(\cancel{\ })$ – those referring to the slack bus – are removed from the computation.
The Jacobian matrix is 4×4.

Assessment of self-understanding

- What is the bus admittance matrix? How many elements it has (for a single-phase equivalent)?
- What does the Jacobian matrix express? How many elements has the Jacobian matrix and why?
- What is the general principle of Newton method?

Future material